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RESEARCH FOCUS

I study how machine learning models can be adapted **efficiently and reliably** when the data they face is degraded, shifted, or adversarial. My work combines parameter-efficient fine-tuning (PEFT) with controlled, diagnostic evaluation — matched protocols, capacity-matched controls, and honest reporting of negative results — and connects it to production AI systems that must keep working under real-world distribution shift.

COMPLETED STUDY — WHEN DOES PEFT HELP UNDER SYNTHETIC DOCUMENT NOISE?

A controlled study of gated PEFT for token-level entity recognition (LayoutLMv3 + LoRA on FUNSD), with every method evaluated under one locked protocol: five seeds, identical optimizer settings, and paired significance tests.

- **A hand-crafted robustness method failed, and is reported as such.** SALLoRA (token quality weighting, layer-wise scalars, gradient sensitivity masking) never beat standard LoRA, degrading macro-F1 by up to 3.09 points under 70% OCR confusion.
- **A learned token gate helps under severe text noise.** GateConcat improves noisy macro-F1 over LoRA by +1.51 points (70% substitution, $p = 0.003$) and +2.30 points (70% OCR confusion, $p = 0.002$).
- **But capacity explains part of the gain.** A constant-gate control (gate fixed at 0.5) captures +0.74 points of the OCR improvement, and GateConcat does not significantly outperform it ($p = 0.21$).
- **The gate is not a corruption detector.** Its detection AUC is about 0.52 (chance level); forcing explicit detection (AUC about 0.91) gives zero downstream benefit, and the gains do not transfer to layout noise (-0.20 points). Robustness claims need capacity controls before mechanism claims.

PRODUCTION AI RESEARCH — LIVENESS & PRESENTATION ATTACK DETECTION

At Jamuna Bank I built a fail-closed face liveness and PAD system (ArcFace matching, Swin-B spatial PAD, 3D ResNet-18 temporal checks, physics-informed replay detection; FastAPI/CUDA inference behind an asyncio gateway, latency cut from over 5 s to about 800 ms). Its deployment became an adaptation study in itself: the initial model rejected 56.25% of confirmed live captures on real phone traffic; a head-only fine-tune failed and was rolled back (53.13%); unfreezing the Swin-B Stage-3 features and fine-tuning on reviewed captures met the pre-registered ship rule on the locked evaluation set — **FAR 0.00% (0/33 replay), FRR 3.125% (1/32 live)** — and was promoted.

DOCTORAL RESEARCH DIRECTIONS

- **Robust & efficient adaptation:** PEFT methods whose benefits survive capacity-matched controls and transfer across corruption types — text, layout, and sensor noise.
- **Trustworthy AI systems:** fail-closed decision design, pre-registered error budgets, and locked evaluation sets as standard research methodology.
- **Uncertainty quantification:** representation-level uncertainty that is demonstrably predictive of failure, tested against null results rather than assumed.

BACKGROUND

Four peer-reviewed papers at IEEE ICCIT 2023 (first-author NLP for mental-health screening; federated learning for medical imaging; PCOS detection; YOLOv8 object detection). RUET Merit Scholar 2020-2022; CGPA 3.30/4.00 with 3.55 over the last 60 credit hours. GRE 304 (Quantitative 164); IELTS Academic 7.0.